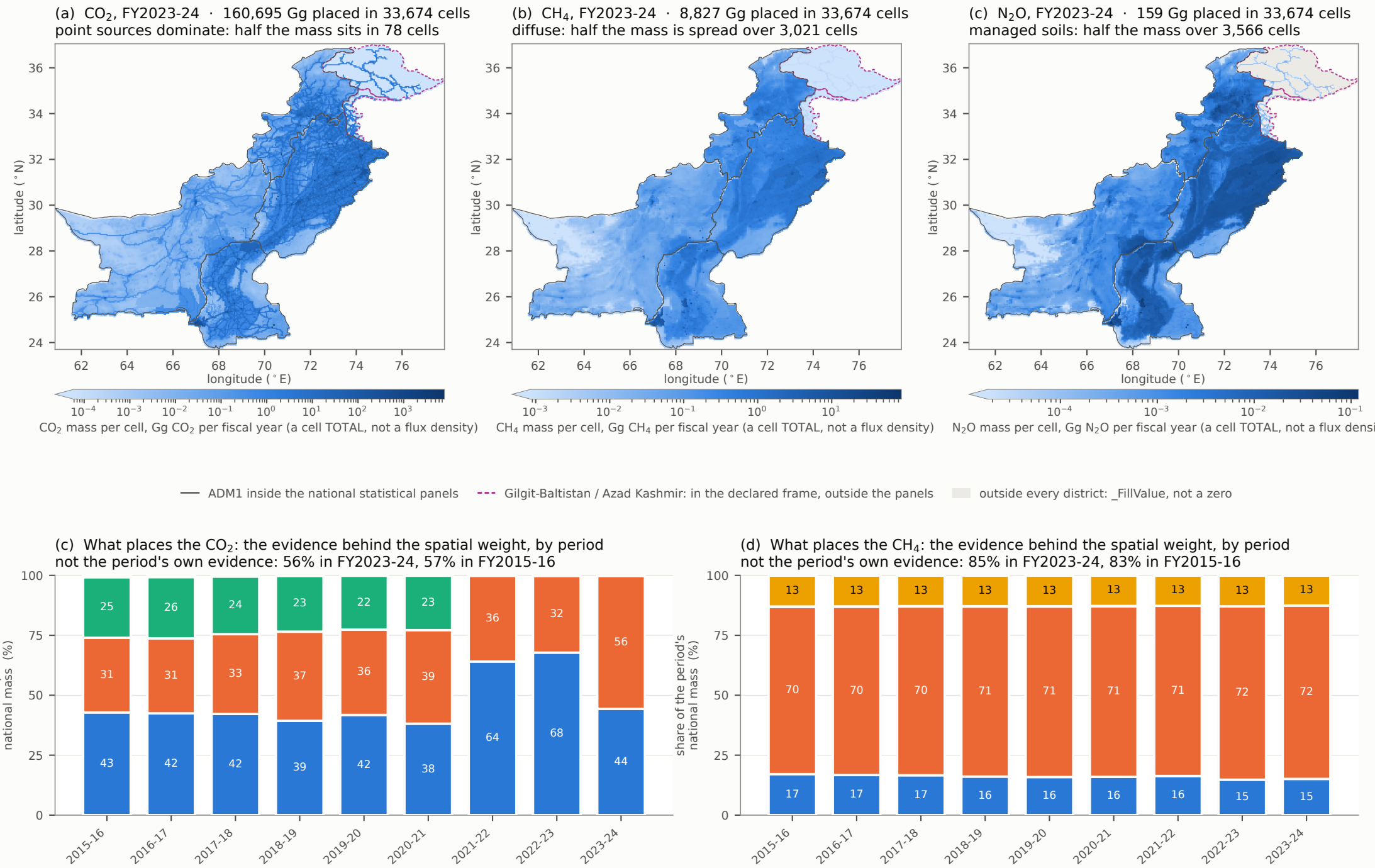
Pakistan greenhouse-gas emissions from the released 0.05° grid, and the evidence that places them

PakEmis 1.0.0rc59 · configuration REF_2023_24 (release pin L 469.63822 Mt CO_2e AR5) · native-gas cell mass in Gg per fiscal year · read from the shipped cube; nothing recomputed



WHAT THIS IS.
Panels (a) and (b) are the released 0.05-degree annual cube for FY2023-24 - release 1.0.0rc59, configuration REF_2023_24 - read, not recomputed. Each cell carries the interval-integrated MASS emitted in it over the fiscal year, in Gg of the NATIVE gas: no GWP is applied, no CO2e is formed, and the value is not divided by cell area, so it is a mass and not a flux density. The cube's three spatial views (physical / origin / accounting) are disjoint and partition the mass, so a cell's value is their sum; dropping one would undercount.

PERIOD.
FY2023-24 is a Pakistani fiscal year, 1 July 2023 to 30 June 2024. It is NOT a calendar year and is not relabelled as one. Panels (c) and (d) run the full nine-period panel, FY2015-16 to FY2023-24.

SOURCES INCLUDED.
Every priced source stream of the 10 estimating modules: agriculture, aviation, fugitives, industry, ippu, power, railways, residential, transport, waste. F-gases, lime (2.A.2), LULUCF and commercial / institutional buildings are not estimated by this release and are therefore absent from the map. Their absence is a coverage gap, not a zero, and the map carries no evidence about them.

THE MASS NOT ON THE MAP.
249.46 Gg CO2 (0.251 Mt CO2e AR5; 0.15% of national CO2) is domestic-aviation cruise, which has no native geometry below the provincial control and is held at ADM1. It is carried as the fifth stacked class in (c) and (d) rather than spread to make the map close. The identity grid + unallocated = national stream ledger holds for all 27 period x gas combinations; worst residual 0.111 ppm of the ledger, which is float32 storage and not a disagreement.

PROJECTION AND AREA CONVENTION.
EPSG:4326 (WGS 84), plotted plate carree at equal aspect; no reprojection. The lattice is equal-ANGLE, so a cell's geodesic area runs from 24.678 km2 at the northern edge to 28.298 km2 at the southern. Because the plotted quantity is a per-cell mass and not a density, that variation does not bias the colour - it varies only the apparent size of a cell. Cells outside every district are _FillValue (-9999) and are drawn as the grey footprint; an in-country zero would be a measurement and would be stored as 0. There are none this period: all 33,674 in-country cells are strictly positive for both gases.

BOUNDARY FRAME.
geoBoundaries gbOpen PAK ADM1/ADM2, represented year 2019, release commit 9469f09 (Runfola et al. (2020), PLOS ONE 15(4), e0231866; Public Domain). The declared frame INCLUDES the 11 territory units of Gilgit-Baltistan and Azad Kashmir, drawn dashed: they are inside the gridded frame but outside the national statistical panels, and they are omitted or drawn differently in several comparator products, so a cell-by-cell comparison against those products is not like-for-like there. Districts carry the 2023 census vocabulary on 2019 geometry; the eleven districts carved out during the window have no contemporaneous boundary, and that vintage mismatch is declared rather than corrected.

WHAT THE MAP DOES NOT SHOW.
No cell is an observation. Panels (c)-(e) are here so that the detail in (a) and (b) is not read as observed local structure. In the reference year 56% of CO2 and 85% of CH4 are placed by a weight that is not the period's own evidence; in FY2015-16 that is 57% and 83%. Every raster surface in (e) predates the fiscal year it places. Per-cell uncertainty is deliberately not stored: within a category the parametric error is perfectly correlated, so a per-cell sigma would invite quadrature and understate every aggregate.